

FOSD — Unit 4

Deadlocks

GATE CSE PYQ source-page collection • 2017–2026

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Included source pages

CS2018 — source page 16 — selected Q39
CS2019 — source page 13 — selected Q39
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- Q.39 In a system, there are three types of resources: E , F and G . Four processes P_0 , P_1 , P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $\text{Max}[P_2, F]$ is the maximum number of instances of F that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

Allocation				Max			
	E	F	G		E	F	G
P_0	1	0	1	P_0	4	3	1
P_1	1	1	2	P_1	2	1	4
P_2	1	0	3	P_2	1	3	3
P_3	2	0	0	P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

- (A) The system is in *safe* state.
- (B) The system is not in *safe* state, but would be *safe* if one more instance of E were available
- (C) The system is not in *safe* state, but would be *safe* if one more instance of F were available
- (D) The system is not in *safe* state, but would be *safe* if one more instance of G were available

- Q.39 Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

- (A) $X_p + X_q < \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (B) $X_p + X_q < \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (C) $\text{Min}(X_p, X_q) \geq \text{Min} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
 (D) $\text{Min}(X_p, X_q) \leq \text{Max} \{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

- Q.40 Consider the following statements:

- I. The smallest element in a max-heap is always at a leaf node
 II. The second largest element in a max-heap is always a child of the root node
 III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time
 IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

- (A) I, II and III (B) I, II and IV (C) I, III and IV (D) II, III and IV

- Q.41 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

- Q.42 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

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Q.25	Consider the following three relations in a relational database. $Employee(\underline{eId}, Name)$, $Brand(\underline{bId}, bName)$, $Own(\underline{eId}, \underline{bId})$ Which of the following relational algebra expressions return the set of $eIds$ who own all the brands?
(A)	$\Pi_{eId}(\Pi_{eId, bId}(Own) / \Pi_{bId}(Brand))$
(B)	$\Pi_{eId}(Own) - \Pi_{eId}((\Pi_{eId}(Own) \times \Pi_{bId}(Brand)) - \Pi_{eId, bId}(Own))$
(C)	$\Pi_{eId}(\Pi_{eId, bId}(Own) / \Pi_{bId}(Own))$
(D)	$\Pi_{eId}((\Pi_{eId}(Own) \times \Pi_{bId}(Own)) / \Pi_{bId}(Brand))$
Q.26	Which of the following statements is/are TRUE with respect to deadlocks?
(A)	Circular wait is a necessary condition for the formation of deadlock.
(B)	In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.
(C)	If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur.
(D)	In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in deadlock state.

Q.48	$P = \{P_1, P_2, P_3, P_4\}$ consists of all active processes in an operating system. $R = \{R_1, R_2, R_3, R_4\}$ consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: $R_1 \rightarrow P_1, R_2 \rightarrow P_2, R_3 \rightarrow P_3, R_4 \rightarrow P_4$ (the assignment edge $R_1 \rightarrow P_1$ means resource R_1 is assigned to process P_1, and so on for others) Claim edges: $P_1 \rightarrow R_2, P_2 \rightarrow R_3, P_3 \rightarrow R_1, P_2 \rightarrow R_4, P_4 \rightarrow R_2$ (the claim edge $P_1 \rightarrow R_2$ means process P_1 is waiting for resource R_2, and so on for others) Which of the following statement(s) is/are CORRECT?
(A)	Aborting P_1 makes the system deadlock free.
(B)	Aborting P_3 makes the system deadlock free.
(C)	Aborting P_2 makes the system deadlock free.
(D)	Aborting P_1 and P_4 makes the system deadlock free.

Q.29	With respect to deadlocks in an operating system, which of the following statements is/are FALSE?
(A)	Banker's algorithm is used to prevent deadlocks
(B)	Deadlock formation can be prevented by ensuring that the hold and wait condition is not allowed
(C)	An assignment edge in a resource allocation graph is marked from a process to a resource
(D)	A safe state guarantees that all processes can finish without formation of a deadlock
Q.30	Let P, Q, R and S be the attributes of a relation in a relational schema. Let $X \rightarrow Y$ indicate functional dependency in the context of a relational database, where $X, Y \subseteq \{P, Q, R, S\}$. Which of the following options is/are always true?
(A)	If $(\{P, Q\} \rightarrow \{R\} \text{ and } \{P\} \rightarrow \{R\})$, then $\{Q\} \rightarrow \{R\}$
(B)	If $\{P, Q\} \rightarrow \{R\}$, then $(\{P\} \rightarrow \{R\} \text{ or } \{Q\} \rightarrow \{R\})$
(C)	If $(\{P\} \rightarrow \{R\} \text{ and } \{Q\} \rightarrow \{S\})$, then $\{P, Q\} \rightarrow \{R, S\}$
(D)	If $\{P\} \rightarrow \{R\}$, then $\{P, Q\} \rightarrow \{R\}$

Q.34	Consider the following program in C: <pre> #include <stdio.h> void func(int i, int j) { if(i < j) { int i = 0; while (i < 10) { j += 2; i++; } } printf("%d", i); } int main() { int i = 9, j = 10; func(i, j); return 0; } </pre> The output of the program is _____. (<i>answer in integer</i>) <i>Note:</i> Assume that the program compiles and runs successfully.
Q.35	Consider a system consisting of k instances of a resource R, being shared by 5 processes. Assume that each process requires a maximum of two instances of resource R and a process can request or release only one instance at a time. Further, a process can request the second instance of the resource only after acquiring the first instance. The minimum value of k for the system to be deadlock-free is _____. (<i>answer in integer</i>)